POTTI SRIRAMULU CHALAVADI MALLIKARJUNA RAO
COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)
APPROVED BY AICTE, NEW DELHI, AFFILIATED TO JNTU KAKINADA

Department of Computer Science Engineering Artificial Intelligence

Final year project proposal for REVIEW-1

Topics chosen are:

1.AI-Powered Face Recognition-Based Attendance Monitoring System
2.AI-Powered Clinical Trial Matcher
3.AI-Powered Personalized Workout Coach
4.Multi-Modal AI Search Engine (Text + Image + Voice + Gesture)

Presented by:

A LOHITH VENKATA GUPTHA -22KT1A4335
MD SHAHINA TABASSUM -22KT1A4320
P DYUTHILAYA -22KT1A4325
V VISHNU VARDHAN -22KT1A4358

TITLE-1:AI-Powered Face Recognition-Based Attendance Monitoring System

Abstract:

Traditional attendance systems—such as manual registers, RFID cards, or biometric scanners—are often inefficient, vulnerable to proxy manipulation, and raise hygiene concerns, especially in shared or high-traffic environments. These conventional methods require physical contact or manual operation, which can be time-consuming and error-prone. To address these limitations, this project introduces an AI-powered Face Recognition Attendance System that leverages advanced computer vision and deep learning techniques to provide a secure, contactless, and real-time attendance solution.The proposed system integrates convolutional neural networks (CNNs) with pretrained deep feature extractors such as FaceNet, VGGFace, or ArcFace to generate high-dimensional facial embeddings. These embeddings capture unique facial characteristics, allowing the system to recognize individuals across a wide range of lighting conditions, facial expressions, orientations, and occlusions. The embeddings are then matched against a secure, encrypted backend database, ensuring seamless identification and attendance logging. Key functionalities include multi-face recognition (identifying multiple individuals in a single frame), contactless logging, anomaly detection (flagging unknown or suspicious entries), and automated attendance reports, which can be periodically emailed or exported in multiple formats (CSV, PDF, etc.). These features collectively reduce human intervention, eliminate proxy attendance, and improve data integrity. To enhance accuracy and adaptability, the system integrates advanced computer vision modules, such as attention mechanisms, facial landmark detection, and bounding box alignment, improving performance even with partial visibility or poor-quality input. It also supports 3D face reconstruction to generate depth-aware facial representations, increasing recognition reliability from non-frontal angles. A notable component of the system is its active learning loop, where the model continuously improves by incorporating newly captured images and feedback. This makes the system self-improving and adaptive over time, catering to evolving user appearances or new additions. For scalability and offline accessibility, the system is designed for edge deployment on devices like Raspberry Pi, NVIDIA Jetson Nano, or Jetson Xavier NX. These embedded AI platforms enable real-time inference and storage without relying on cloud servers, ensuring data privacy and reducing latency—ideal for institutions in bandwidth-constrained environments. Furthermore, multi-modal biometric fusion—which combines facial recognition with voice analysis, gait recognition, and optional QR code fallback—provides additional layers of reliability, inclusivity, and accessibility, especially in edge cases where face data may be insufficient (e.g., wearing masks or low lighting). The system also supports role-based dashboards, allowing administrators, teachers, or HR managers to view, filter, and analyze attendance data through an intuitive user interface. This comprehensive and intelligent attendance system is suitable for schools, colleges, corporate offices, factories, and government facilities, ensuring hygienic, accurate, and secure attendance management. By eliminating human error, reducing administrative workload, and enabling smart analytics, this AI-powered solution represents a significant advancement in the digital transformation of organizational operations.

Keywords:
Convolutional Neural Networks (CNN), Computer Vision, Deep Learning, Facial Embeddings, Edge Computing, Multi-modal Biometrics, Active Learning, 3D Face Reconstruction, Raspberry Pi.

References

[1] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A Unified Embedding for Face Recognition and Clustering," 2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2015, pp. 815–823, doi: 10.1109/CVPR.2015.7298682.
[2] O. M. Parkhi, A. Vedaldi, and A. Zisserman, "Deep Face Recognition," British Machine Vision Conference (BMVC), 2015. [Online]. Available: http://www.robots.ox.ac.uk/~vgg/publications/2015/Parkhi15/parkhi15.pdf.

TITLE-2:AI-Powered Clinical Trial Matcher

Abstract:

Clinical trial recruitment is a critical but resource-intensive process in medical research, often hindered by the complexity of patient data and the intricate eligibility criteria of trials. This project presents an AI-powered Clinical Trial Matcher that addresses these challenges by integrating advanced natural language processing (NLP), machine learning (ML), and computer vision technologies into a unified solution for more efficient and accurate patient-to-trial matching.At the heart of the system is its ability to process and understand unstructured data from electronic health records (EHRs). Using sophisticated NLP models, the system extracts key clinical information such as demographics, medical history, diagnoses, and physician notes. These details are then structured into standardized formats that can be semantically matched with the eligibility criteria of ongoing clinical trials.The system utilizes retrieval-augmented large language models (LLMs) to perform deep semantic matching. These models assess how well a patient's profile aligns with a trial's inclusion and exclusion criteria, generating eligibility scores for each criterion. More importantly, the system provides transparent, interpretable rationales for these scores, enabling clinicians to understand and validate AI-driven recommendations. To further improve the efficiency of matching, the platform incorporates intelligent filtering and similarity-based ranking. This allows the system to prioritize and personalize trial recommendations based on how closely a patient's profile aligns with available trials. By automating and optimizing this process, the system greatly reduces manual effort, enhances the accuracy of matches, and significantly boosts recruitment and enrollment rates. In addition to trial matching, the system enhances site-level operations through the use of computer vision. Facial recognition and timestamp logging are used to automate participant check-ins during trial visits. This not only ensures accurate tracking of visit adherence but also reduces manual data entry errors, enhances data integrity, and improves audit readiness. Such automation minimizes administrative tasks for clinical staff, allowing them to dedicate more time to patient care and oversight. A key innovation of this solution is its multimodal data fusion capability. By combining insights from both NLP-driven EHR analysis and computer vision-based attendance monitoring, the system creates a holistic view of each patient's engagement in the clinical trial. This unified perspective supports ongoing eligibility reassessment, helps adjust protocols dynamically, and informs better decision-making in trial management. Overall, the AI-powered Clinical Trial Matcher transforms the recruitment and operational processes in clinical research. By automating data extraction, enhancing matching precision, and streamlining site-level workflows, the platform improves the scalability, accuracy, and speed of clinical trial execution. It empowers research institutions to recruit the right patients faster, reduce operational burdens, and maintain high standards of data integrity and compliance—ultimately accelerating the development of new therapies and improving patient outcomes.

Keywords:
Clinical Trial Recruitment, Electronic Health Records (EHR), Natural Language Processing (NLP), Large Language Models (LLM), Semantic Matching, Machine Learning (ML), Patient Eligibility Matching, Retrieval-Augmented Generation (RAG), Computer Vision, Facial Recognition, Multimodal Data Fusion,Audit Readiness.

References

[1] J. Luo, H. Wu, P. Zhao, et al., "Deep Learning for Identifying Eligible Patients for Clinical Trials: A Case Study Using Clinical Trial Reports," IEEE Journal of Biomedical and Health Informatics, vol. 25, no. 4, pp. 1026–1037, April 2021. DOI: 10.1109/JBHI.2020.3017887.
[2] S. Rajkomar et al., "Scalable and Accurate Deep Learning with Electronic Health Records," npj Digital Medicine, vol. 1, no. 18, 2018. DOI: 10.1038/s41746-018-0029-1.

TITLE-3: AI-Powered Personalized Workout Coach

Abstract:

With the rising demand for personalized fitness solutions, artificial intelligence (AI) offers transformative capabilities in redefining physical training. This project presents the design and development of an AI-powered personal workout coach that intelligently adapts to individual goals, physiological parameters, and real-time performance metrics. By integrating machine learning, computer vision, and real-time feedback mechanisms, the system delivers customized workout plans, posture correction, and motivational insights to improve user engagement and fitness outcomes. The model processes multi-source data, including input from wearable devices, camera-based form analysis, and direct user feedback, to continuously optimize exercise routines. Wearables such as smartwatches and fitness trackers provide valuable physiological data like heart rate, sleep quality, oxygen saturation, and step count, which the AI uses to adjust workout intensity and type. Supervised learning algorithms recommend exercises tailored to each user profile, while convolutional neural networks (CNNs) enable accurate posture and movement assessment during workouts, minimizing the risk of injury. To ensure natural and intuitive user interaction, natural language processing (NLP) is incorporated, allowing voice-controlled operations. Users can ask for exercise alternatives, receive instructions, and get performance insights through spoken commands. The system is further enhanced with emotion-aware coaching, utilizing facial expression and voice tone analysis to detect user mood and dynamically adjust workout intensity, ensuring better motivation and adherence. Reinforcement learning algorithms are employed to dynamically tailor and evolve fitness plans based on user progress, feedback, and changing health or fitness objectives. Additionally, support for augmented reality (AR) provides live, immersive workout demonstrations via mobile AR or AR glasses, increasing understanding and precision during exercises. A multi-language NLP interface ensures inclusivity by supporting regional and native languages, making the AI coach accessible to users from diverse linguistic backgrounds. Integration with electronic health records (EHRs) allows the system to provide medically-informed fitness plans, especially for users with chronic conditions, prior injuries, or ongoing rehabilitation needs. This ensures safety and personalization for all users, regardless of health status. To further enrich the user experience, the platform includes community engagement and gamification features such as leaderboards, fitness challenges, and virtual workout groups. These elements help maintain motivation and consistency through social interaction and healthy competition. Recognizing the digital divide, the system also supports offline functionality, allowing users in low-connectivity or rural areas to access workout plans and coaching tools without needing constant internet access. Lightweight AI models enable essential features to operate seamlessly offline.

Keywords:
Artificial Intelligence (AI), Personalized Fitness, Workout Recommendation System, Computer Vision, Convolutional Neural Networks (CNNs), Natural Language Processing (NLP), Reinforcement Learning, Augmented Reality (AR), Wearable Devices, Fitness Tracking, Emotion Recognition, Health Informatics, Voice Interactio, Offline AI, Gamification

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References:

[1] M. Zeng, M. S. Bhuiyan, L. Xu, et al., "Convolutional Neural Networks for Human Activity Recognition Using Mobile Sensors," IEEE Transactions on Industrial Informatics, vol. 13, no. 3, pp. 1166–1173, 2017. DOI: 10.1109/TII.2016.2628880.

[2] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A Unified Embedding for Face Recognition and Clustering," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2015, pp. 815–823. DOI: 10.1109/CVPR.2015.7298682.

TITLE-4: Multi-Modal AI Search Engine (Text + Image + Voice + Gesture)

Abstract:

This project presents a Multi-Modal AI Search Engine, in which users can carry out intelligent and natural searches with text, image, voice, and gesture inputs. Most search engines are based on keyword-oriented text inputs, which might be constrained in cases where users might prefer or need different input modalities. Using the newer technologies in Artificial Intelligence (AI), such as Natural Language Processing NLP, CV, Speech Recognition, and Gesture Detection, this platform provides a context-aware, inclusive, and flexible information retrieval experience. For text-based requests, the system uses Transformer-based NLP models like BERT and T5 Text-To-Text Transfer Transformer) to examine and comprehend the semantic meaning of user requests in order to provide more accurate and contextually appropriate search results . For image input, the system uses CNNs and ViT to extract deep features from visual content. To facilitate cross-modal comprehension between images and text, we use CLIP Contrastive Language-Image Pre-training, which projects both modalities into a common embedding space, allowing users to search for images from text descriptions or locate textual information from visual input.For voice search, the search engine incorporates ASR technologies like Wav2Vec2 and Google Speech API, which convert spoken queries to text. These transcriptions are then subjected to the NLP pipeline for accurate semantic matching. Also, the system incorporates gesture-based search functionality, whereby the users are able to execute certain hand or body gestures picked up by a webcam or smartphone camera. The gesture module applies Pose Estimation algorithms such as MediaPipe and OpenPose to detect hand gestures or facial gestures, converting them into command-based or contextual searches, particularly beneficial in accessibility-oriented settings or hands-free interactions. All input modalities are mapped to high-dimensional vectors and indexed by vector similarity search engines such as FAISS or Milvus, facilitating real-time retrieval by semantic proximity instead of keyword matching. Python and Flask are utilized in the backend architecture with connection to MongoDB for metadata storage, whereas the frontend interface is built using HTML, JavaScript, and Web APIs for cross-platform compatibility and responsiveness. This multimodal strategy is useful in many areas like e-commerce, education, healthcare, and digital libraries, where users might not necessarily have syntactically structured queries or require visual or voice-based interaction. A student, for instance, can submit a diagram to look for relevant subjects, a patient can provide symptoms through voice, or a visually impaired user can interact through gestures.Even with its prowess, the system is challenged by high computational requirements, generalization problems across various domains, and the inability to deal with ambiguous or multi-intent inputs. Planned improvements include support for multiple languages, gesture adaptation, incorporation of Large Language Models (LLMs) such as GPT for generative responses, and ranking using reinforcement learning to make results more relevant with time. Improving on privacy and security, especially from voice and image inputs, also is an important consideration. Overall, the outlined Multi-Modal AI Search Engine is a remarkable advancement in human-computer interaction, providing users with an efficient, available, and natural means of accessing the virtual world using several modes of input that suit their requirement and inclination.

Keywords

Multimodal Search, Artificial Intelligence, Semantic Retrieval, Natural Language Processing (NLP), Computer Vision (CV), Speech Recognition, Gesture Recognition, CLIP, BERT, T5, Wav2Vec2, Vision Transformers, Vector Similarity Search, Cross-Modal Retrieval, Human-Computer Interaction, Automatic Speech Recognition(ASR), Voice-Based Search.

References

[1] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. Proceedings of NAACL-HLT, pp. 4171–4186.

[2] Radford, A., Kim, J. W., Hallacy, C., et al. (2021). *Learning Transferable Visual Models From Natural Language Supervision*. Proceedings of the 38th International Conference on Machine Learning (ICML).